Stats → LA → Analysis → Python & Pandas + Numpy + ~

↳ Machine learning section

⇓

eg. D_x

classification

'Analysis' ← D_x

Historical

id	income	credit-score	status (Loan)
c1	14	650	Yes
c2	15	680	Yes
c3	7	450	No
c4	8	500	No

→ ML()

c6

↳ Build

c_5 → [9.5, 600] → Yes / No → How?

known

unknown → predict

→ if income > 10 and CS > 500

→ Yes

else

→ No

⇑

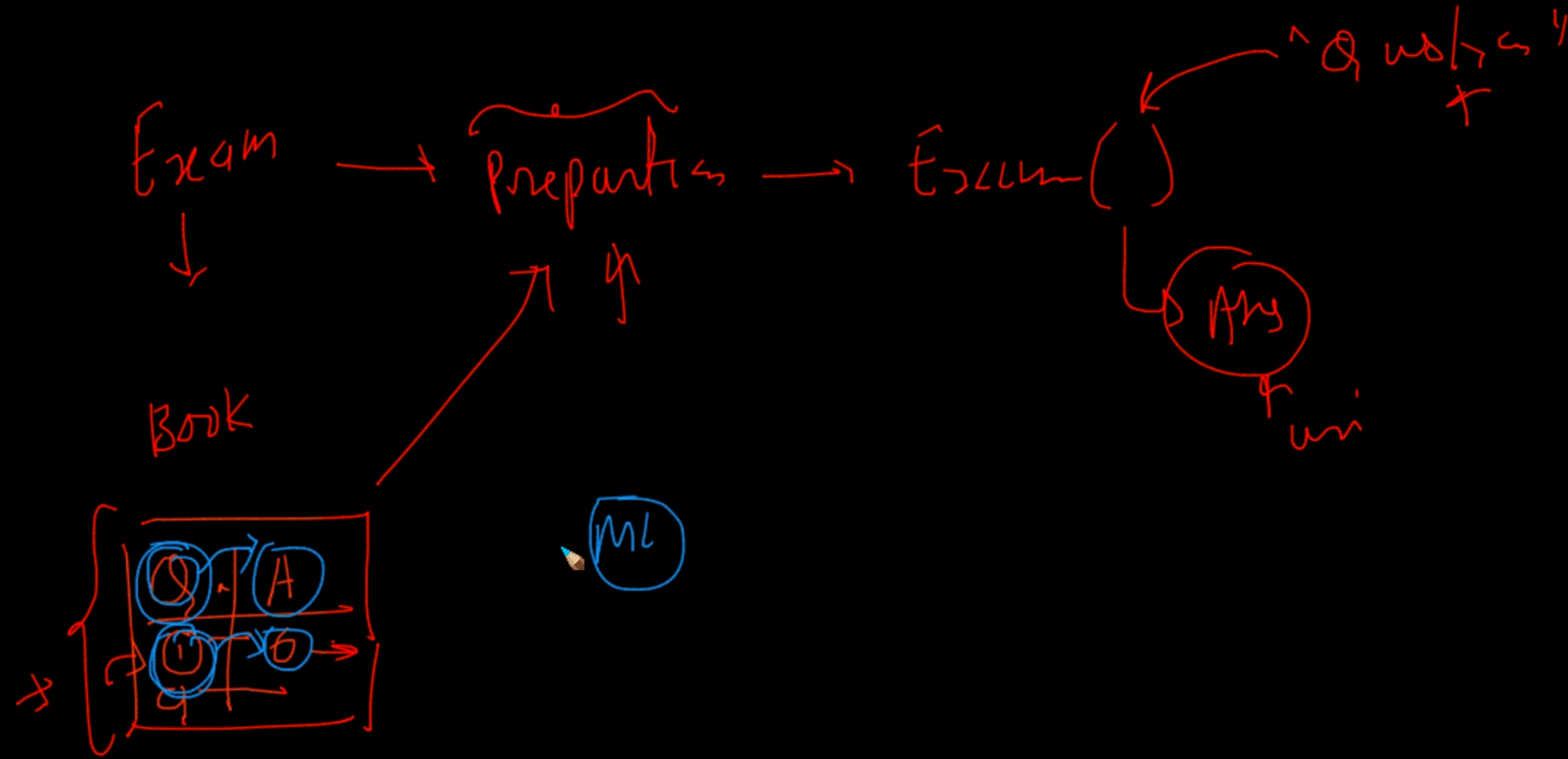
Rule

∈ [Yes, No]

D (X, Y)

→ M() → Yes / No

X


Exam
Preparation
Exam
"Questions"
Ans
Book
Q
A
ML

Decide

2 input →

income > 10 and CS > 500

→ yes

else

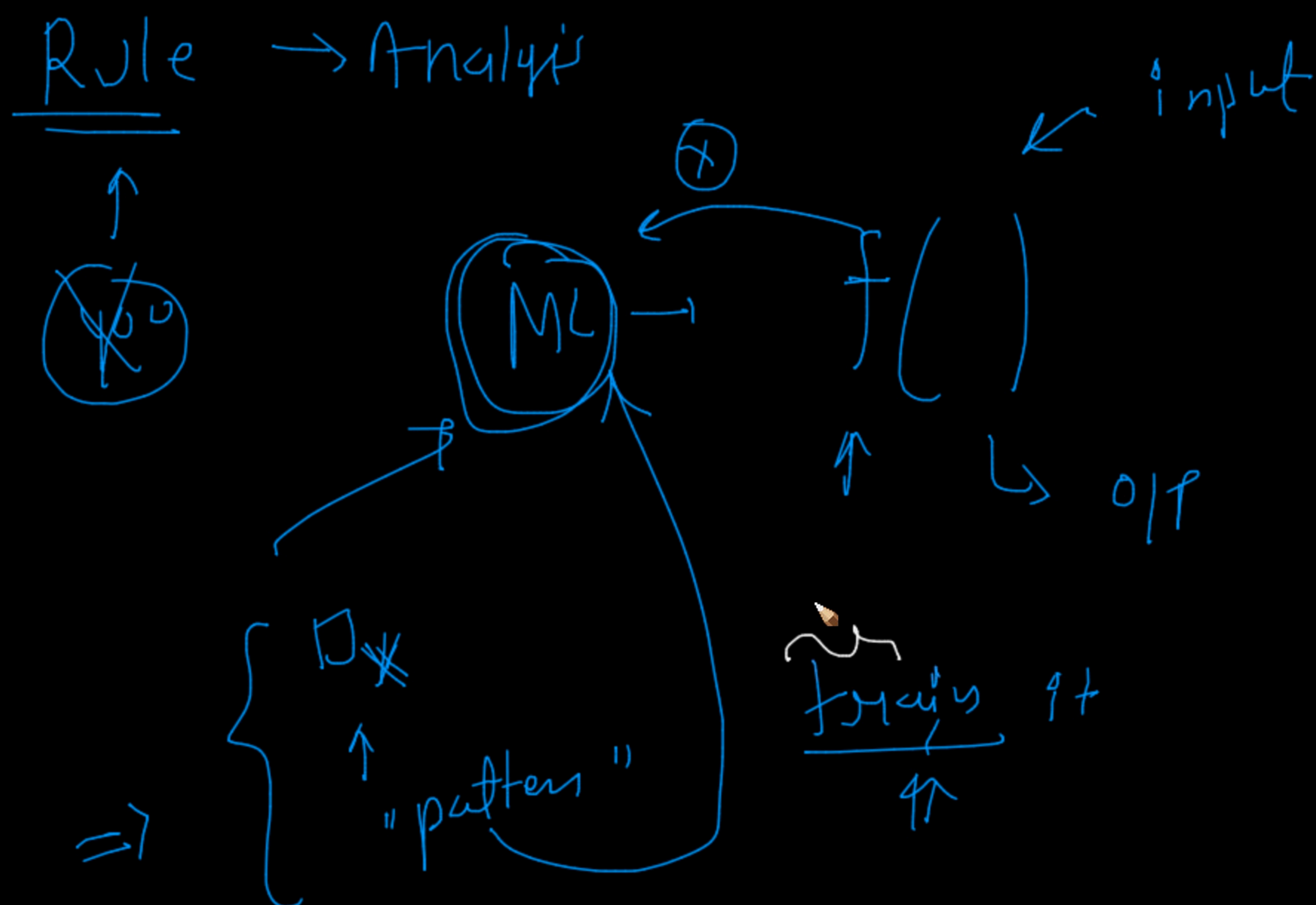
NO

'Rule'

f_1	f_2	f_3	Y
–	–	–	

3 information

Analysis


Rule → Analysis
input
ML
f( )
O/P
Training it
"pattern"
D

what kind of data you have

for eg.

ML

Supervised ML

Unsupervised ML

what kind of data you have

for eg.

labelled data

$D_x(\bar{X}, \bar{Y})$

$M(\ )$

Y

X

ML

Supervised ML

Unsupervised ML

'Clustering'

" K-means "

$O(X) \rightarrow M()$

Classification

(X, Y)

$\in [- - - -]$

set of value

Regression Algo

Y

14

$M'(\ )$

Real

$M(x_1)$

Real

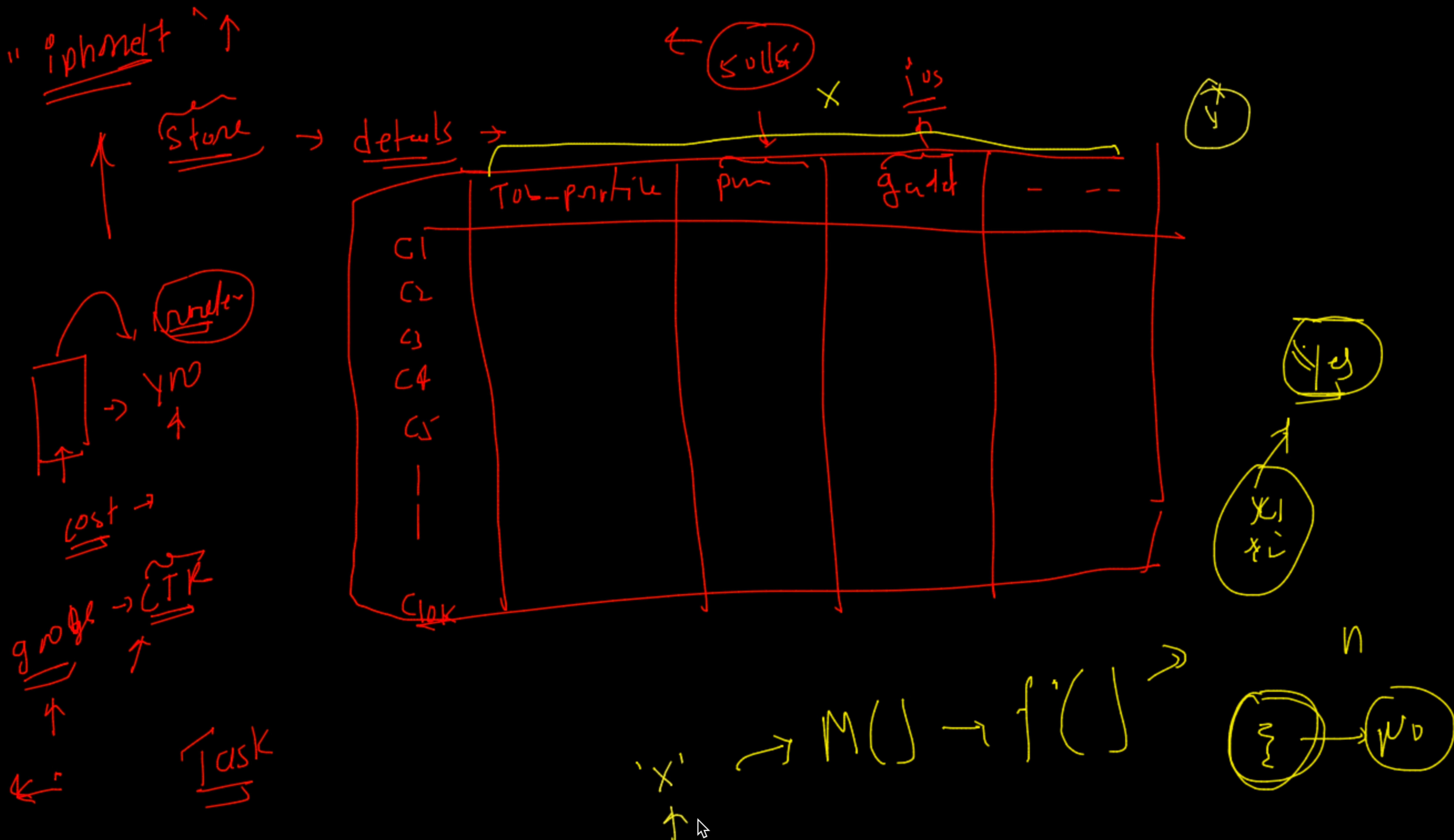
eg

D

X

Y

Discount	Sales
10%	300
20%	400
12	320
15%	380

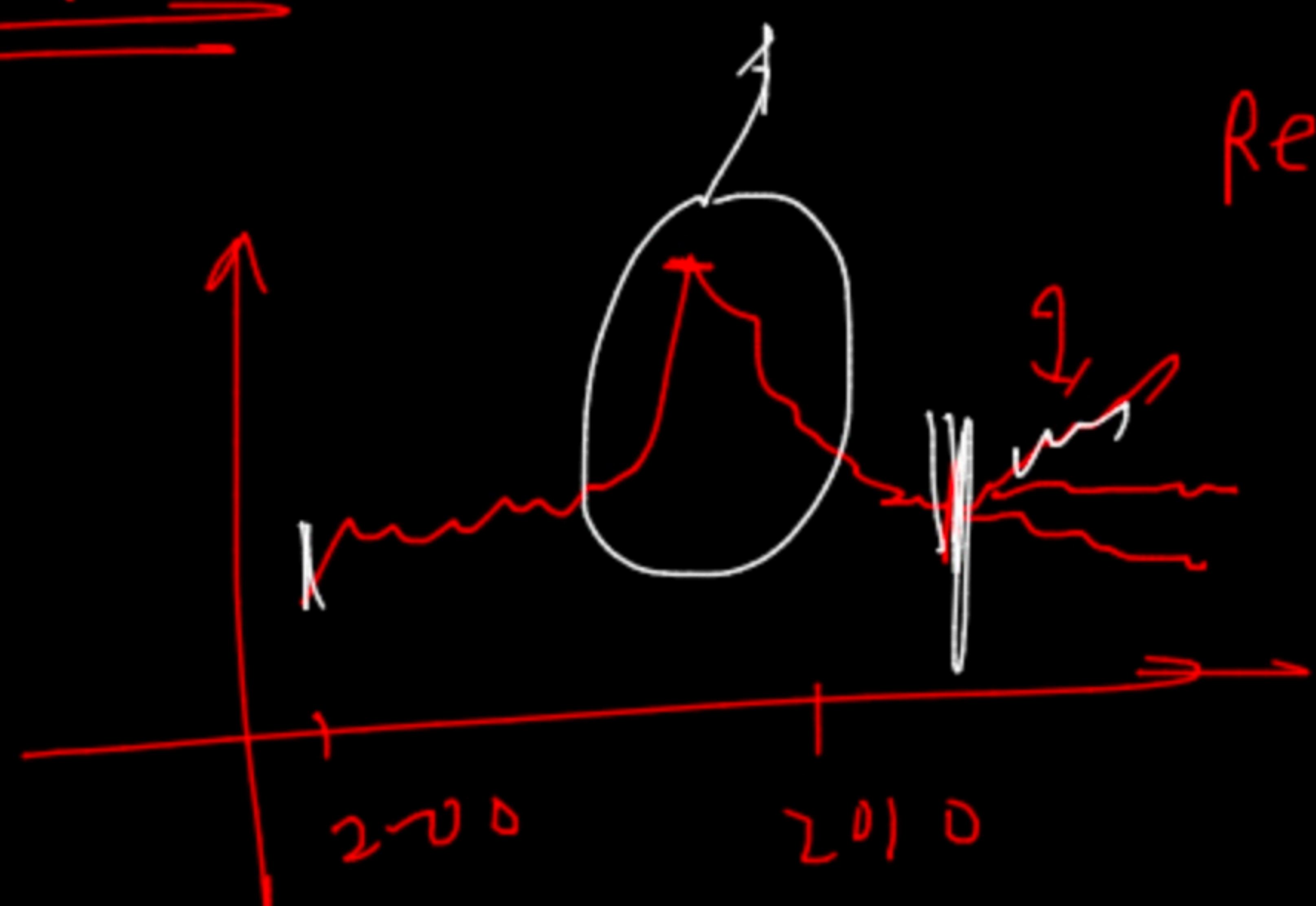

iphone7
store
details
ios
pin
C1
C2
C3
C4
C5
C10K
y/no
cost
google
CTR
Task
'x'
M( ) → f( )
yes
no
n

7 / 11

salary
Data → Hiring → (syn)

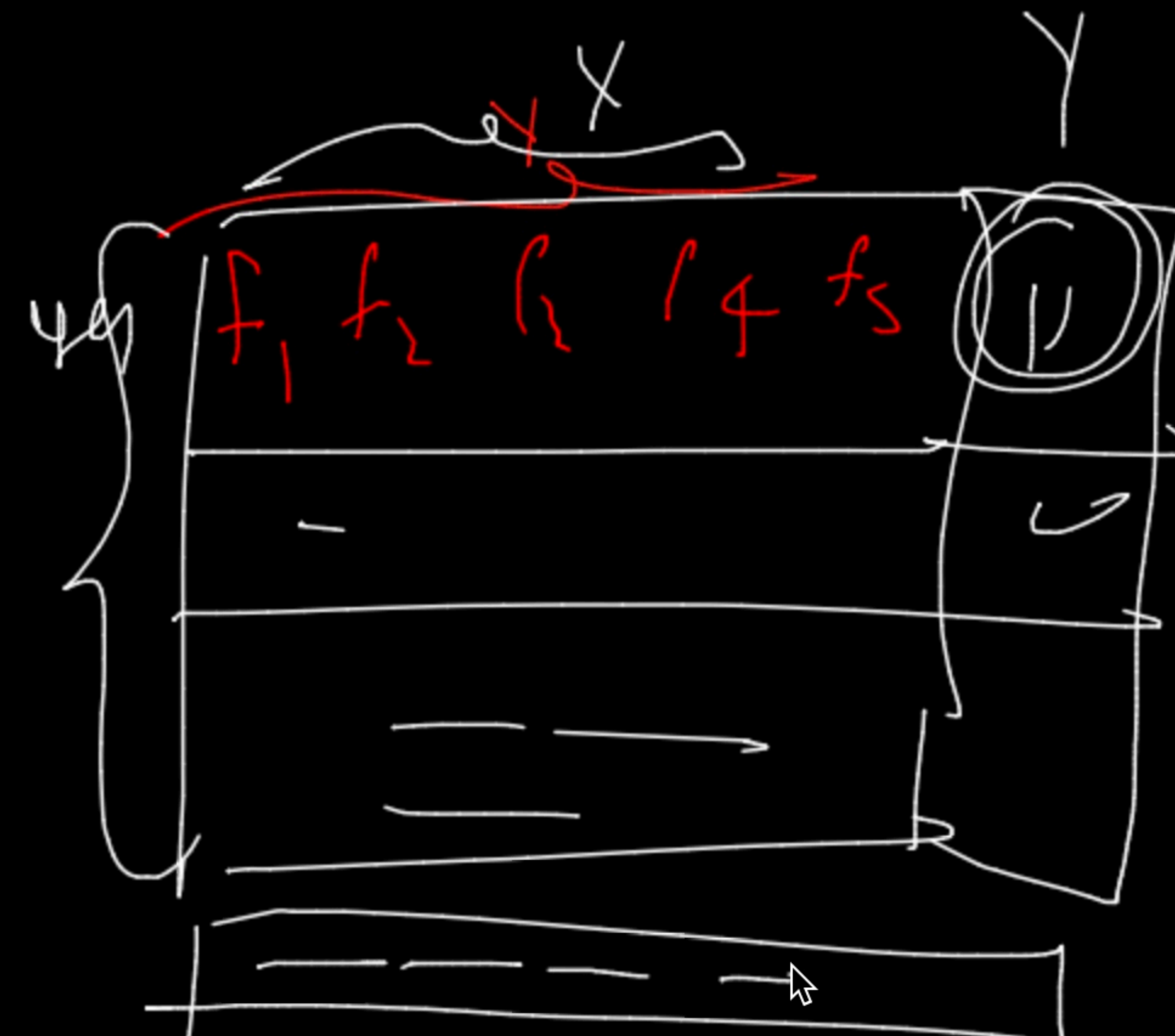
Projections

more → ? → factors →



Real

factors

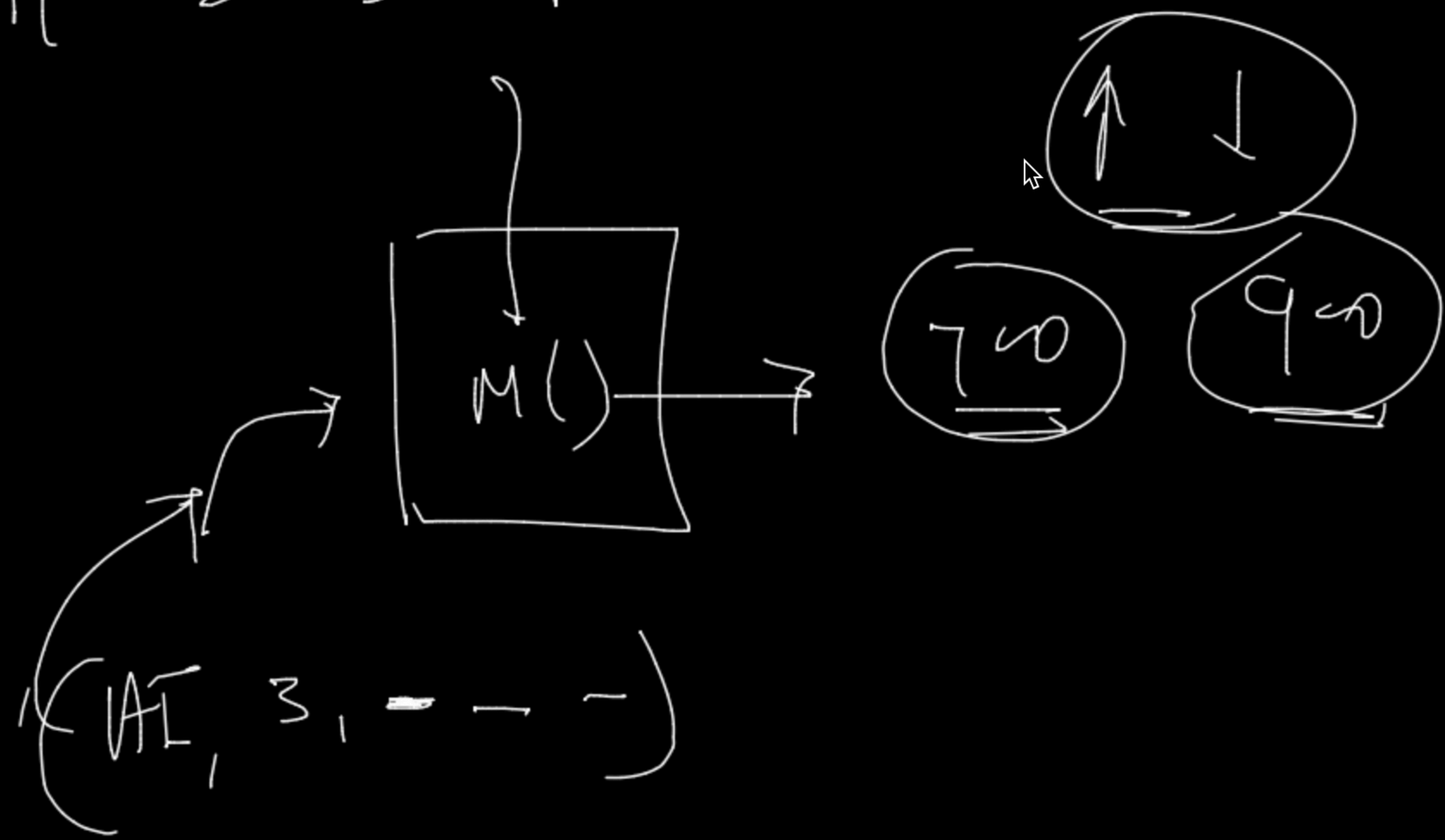


labelled

Regression

History

f_1 f_2 f_3



(smile)
Simple → Classification

simple (smile) → classification

"KNN" — K nearest neighbours

parameters

eg.

80 %

D → train (X train , Y train)

Random split

20 % D test (X test , Y test)

S' → [ ] → 50 %

→ $\boxed{50\%}$

$\frac{1}{2}$

I	CS	S	pred
14	650	Y	N
15	780	Yes	Yes

eg. D →

income	CS	Status
14	650	Y
15	750	Y
6	360	N
5	340	N
15	780	Yes

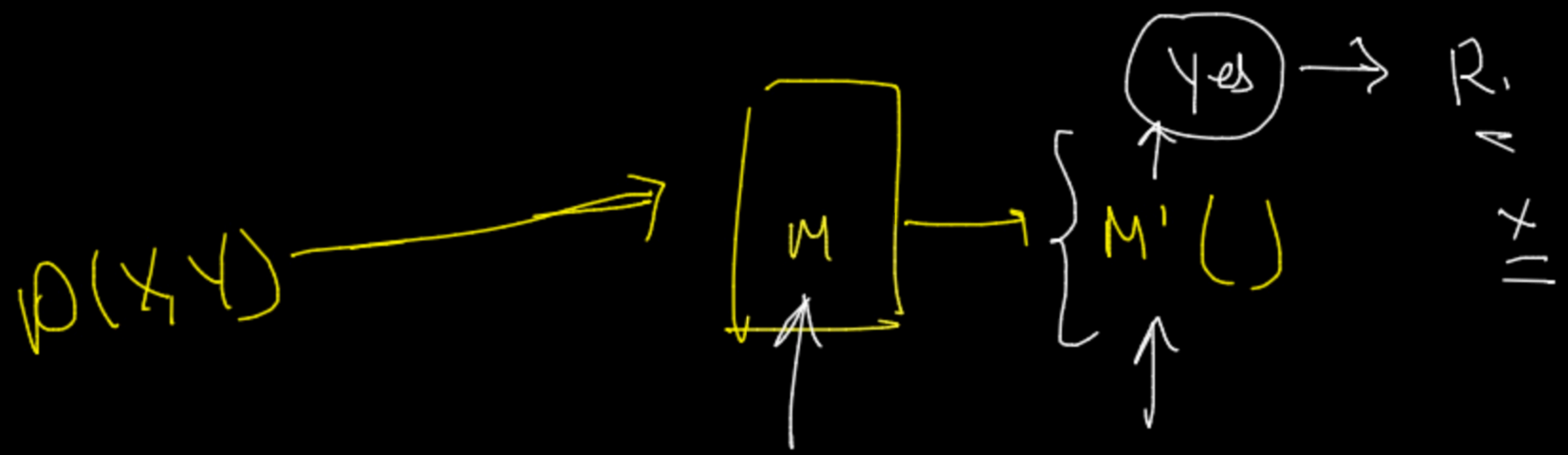
Historical

X Y

I	CS	S
15	750	Y
6	350	N
5	340	N
~~15~~	~~780~~	~~Y~~

→ M( ) → $\boxed{N}$

Y



testing → future

KNN Algorithm

parameter

any integer

$$D_x \rightarrow \begin{cases} D_{Train}(X_{train}, Y_{train}) \\ D_{Test}(X_{test}, Y_{test}) \end{cases}$$

$D_{train} \rightarrow$

	income	cs	status
P1	14	750	yes
P2	15	780	yes
P3	3	350	N
P4	6	880	N

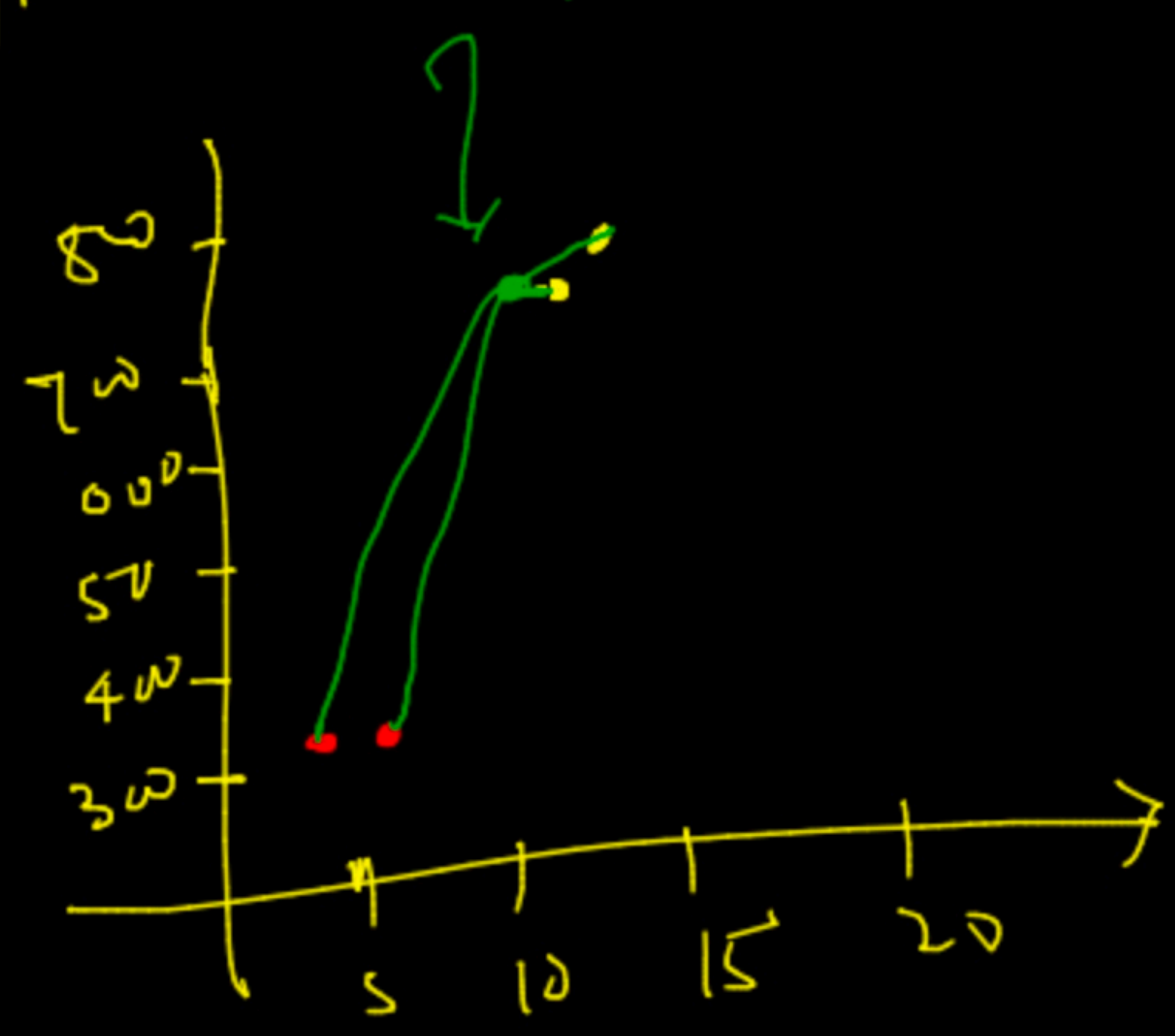
1 crore

income	cs	status
13	750	Yes
3	360	No

it will → memory

training data → memory

at time of training



step → distance

X_{query} (13, 750), (14, 75)

distance → to ∀ training pts

$(X_q, P_1) = \sqrt{(14-13)^2 + (0)^2}$
$\Rightarrow 1$

$(X_{q_1}, P_2) = 1.04$

$(X_{q_1}, P_3) = 3.4$

step 2 at time of testing → D_{Test}

$(X_{q_1}, P_4) = 3.6$

sorting → increasing →

$(X_{query}, P_1) = 1$

$(X_{query}, P_2) = 1.04$

$(X_{query}, P_3) = 3.64$

$(X_{query}, P_4) = 3.46$

training pt

$\boxed{k=3}$

k=?

k=2

k=3

P_1 → Yes

P_2 → Yes

P_4 → No

P_3

Yes → 2
No → 0

⇓

decide

⇓

Yes → 2
No → 1

→ Yes

X_{query} → $\boxed{\text{Yes}}$

predicting

k=n → find out

K = 1

65% 75% 85% 94% 88%

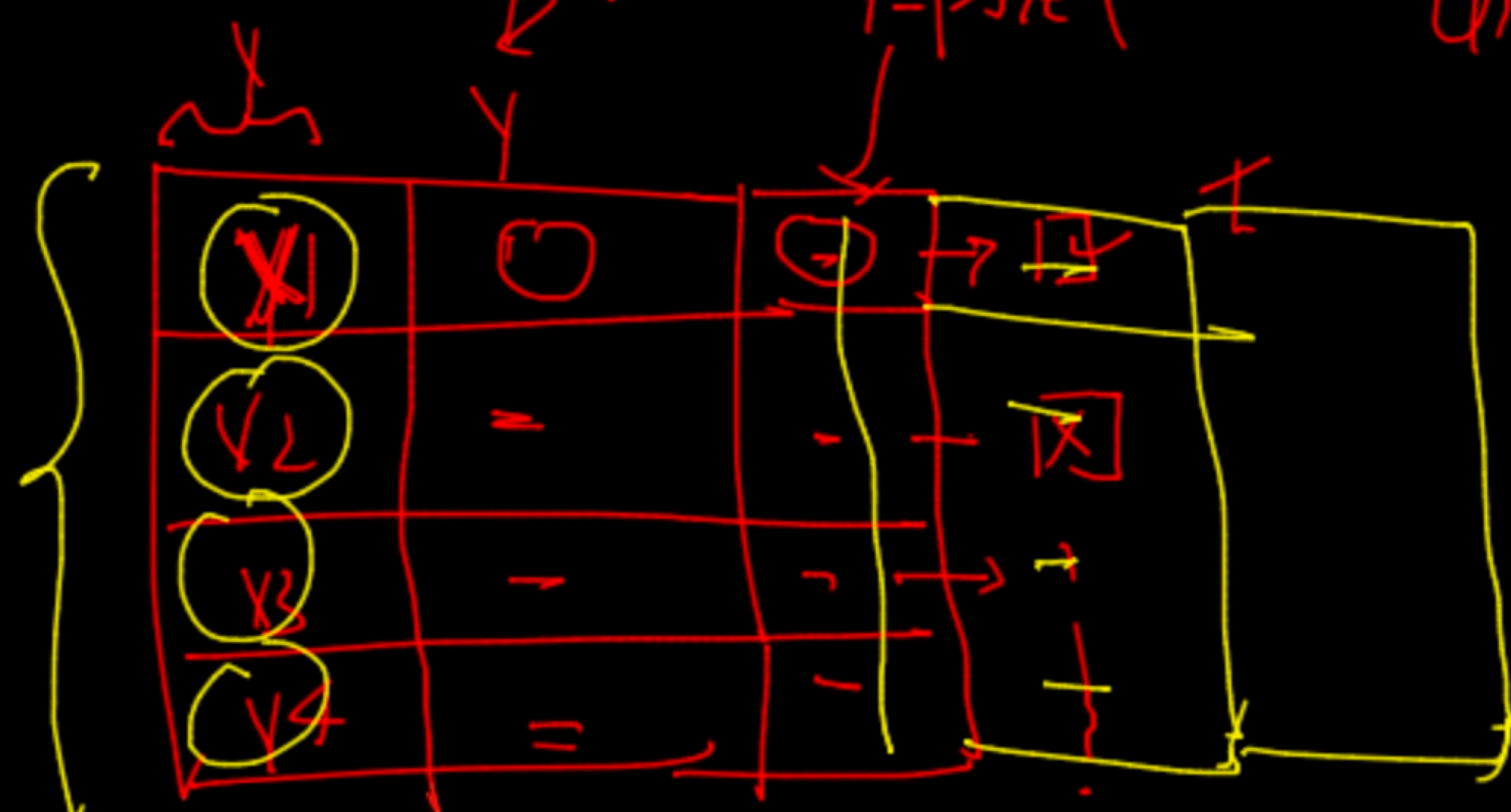
M(K=1) → prediction

actual

y-pred

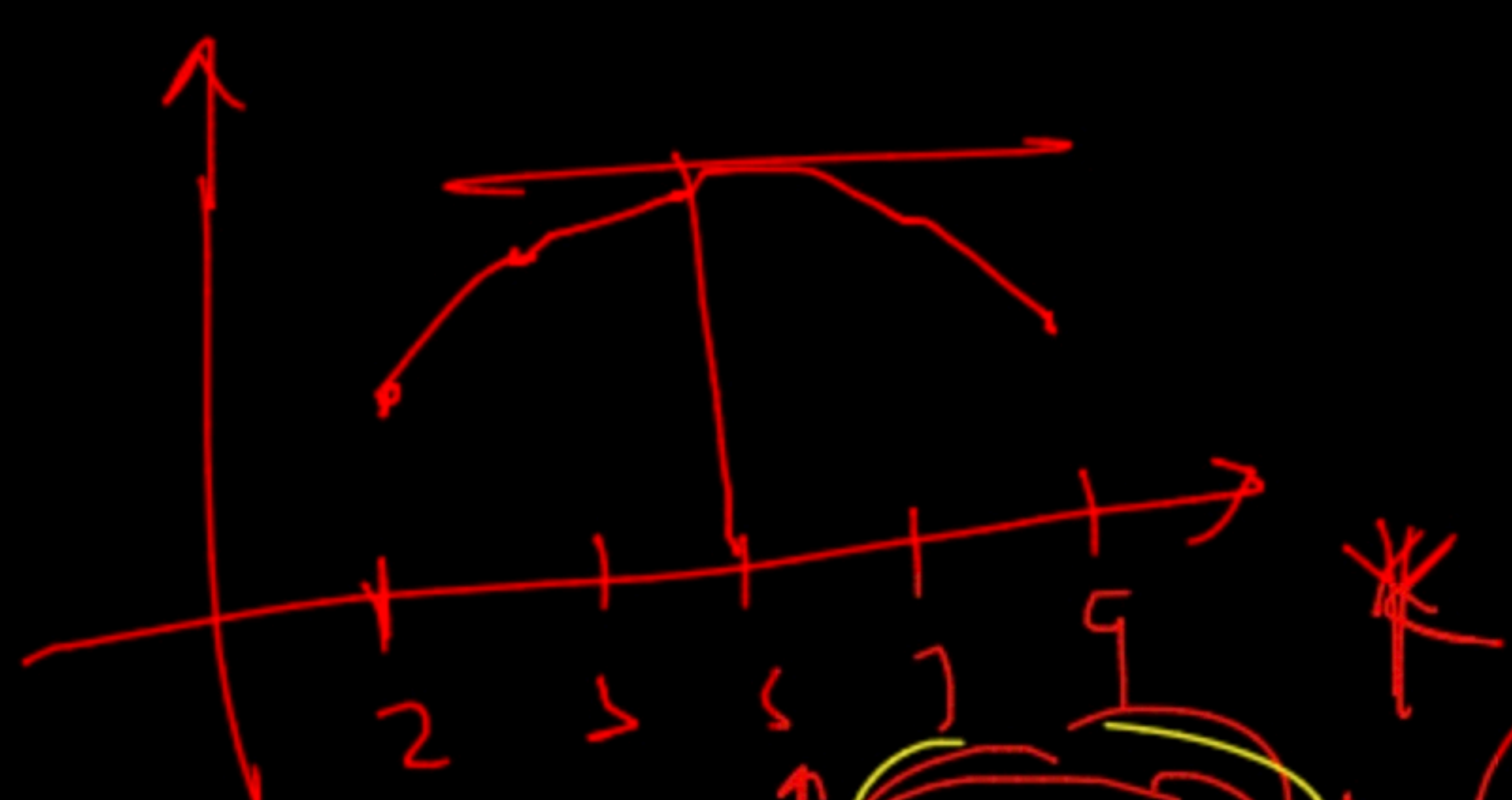
dropped

X1



Exam

90%



'Data leakage'

Acc = $\frac{3}{4}$

Book (Q–N)

(Q–N)

(Q–N)

50%

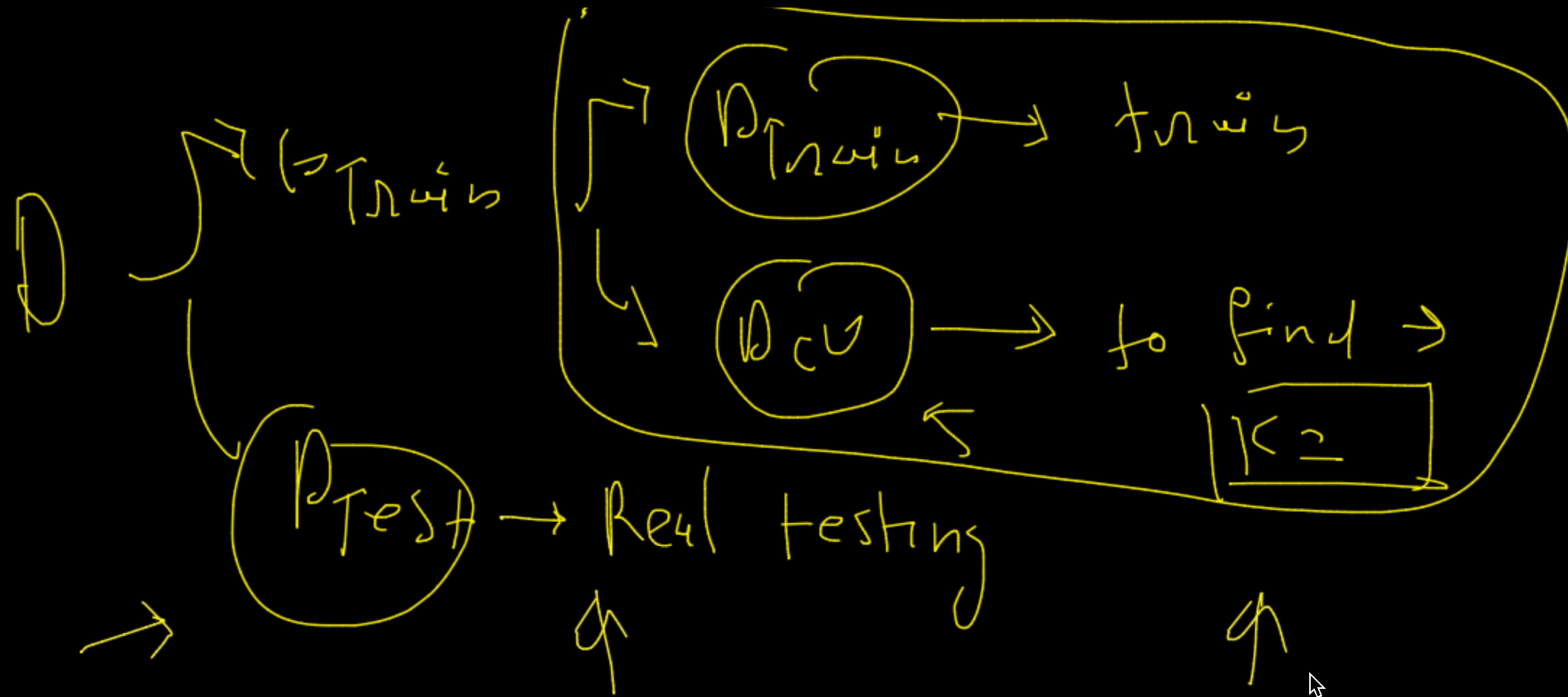
40%

K=5

Best

Result

16 / 16


D
D Train
D Train
train
D CV
to find
K =
D Test
Real testing